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To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.	1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities. 2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program. 3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

Experiment No. 06

Aim: - Write a java program for Method overloading and Constructor overloading.

Software Requirement: - Java Development Kit (JDK), Integrated Development Environment (IDE) (NetBeans), Java Runtime Environment (JRE).

Theory: -

a) Method Overloading

- Method overloading in Java is the feature that enables defining several methods in a class having the same name but with different parameters lists.
- These algorithms may vary with regard to the number or type of parameters.
- When a method is called, Java decides which version of it to execute depending on the arguments given.
- If we have to perform only one operation, having the same name of the methods increases the readability of the program.
- Method overloading in Java is also known as Compile-time Polymorphism, Static Polymorphism, or Early binding. In Method overloading compared to the parent argument, the child argument will get the highest priority.
- Suppose you have to perform the addition of the given numbers, but there can be any number of arguments if you write the method such as a(int,int) for two parameters, and b(int,int,int) for three parameters then it may be difficult for you as well as other programmers to understand the behavior of the method because its name differs.

Syntax:

```
class ClassName {
   // Method with one parameter
    returnType methodName(parameterType1 parameterName1) {
       // Method body
   }

   // Overloaded method with a different number of parameters
   returnType methodName(parameterType1 parameterName1, parameterType2 parameterName2) {
       // Method body
   }

   // Overloaded method with the same number but different types of parameters
```

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```
    returnType methodName(parameterType1 parameterName1, parameterType2 parameterName2) {
        // Method body
    }

    // Overloaded method with parameters in a different order
    returnType methodName(parameterType2 parameterName2, parameterType1 parameterName1) {
        // Method body
    }
}
```

Example:

```
int myMethod(int x)
float myMethod(float x)
double myMethod(double x, double y)
```

Different Ways of Method Overloading in Java

- Changing the Number of Parameters.
- Changing Data Types of the Arguments.
- Changing the Order of the Parameters of Methods

Advantages of Method Overloading

- Method overloading improves the Readability and reusability of the program.
- Method overloading reduces the complexity of the program.
- Using method overloading, programmers can perform a task efficiently and effectively.
- Using method overloading, it is possible to access methods performing related functions with slightly different arguments and types.
- Objects of a class can also be initialized in different ways using the constructors.

b) Constructor Overloading

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- Just like methods are overloaded in Java, overloading of constructors is also possible.
- We can define multiple constructors within a class and each of them can have different types of parameters.
- With the help of constructor overloading, objects of the class can be created with different kinds of information, we want to add to it.
- Constructor Overloading helps in creating multiple constructors within a class, each accepting different parameters. This can be useful in many ways such as,
  -it provides flexibility in object creation.
  -it enables you to initialize objects in different ways depending on the information available or needed at the time of creation.
  -it also helps in making the code more simple.

Syntax

```
public class MyClass {

  // Constructor with no parameters
  public MyClass() {
     // Initialization code
  }


  // Constructor with one parameter
  public MyClass(int parameter) {
     // Initialization code using the provided parameter
  }


  // Constructor with multiple parameters
  public MyClass(int parameter1, String parameter2) {
     // Initialization code using the provided parameters
  }


  // Additional constructors as needed...
}
```

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Benefits of Constructor Overloading in Java

Constructor Overloading offers several benefits, some of its key advantages are as follows:

- **Flexibility-**Constructor Overloading allows developers to create objects in different ways by providing multiple constructors with different parameter lists making the process of initializing objects more flexible.
- **Code Reusability-** Insteadof duplicating code or creating similar constructors with minor variations, developers can overload existing constructors to accommodate new initialization requirements without redundancy.
- **Code Readability-** While overloading constructors, developers can provide meaningful names to constructors and parameters that will convey the purpose and usage of each constructor effectively, enhancing the code readability and understanding.
- **Maintainability-** With constructor overloading, developers can easily create new or modify existing constructors as they want to. It won't affect the main codebase in any way.
- **Enables Polymorphic Behavior-** Constructor Overloading also contributes to the **polymorphism in Java** classes, enabling objects to exhibit different behaviors based on the parameters used to instantiate them.

Program and Output:

a) Method Overloading

```
class Calculator {
    // Method to add two integers
    public int add(int a, int b) {
        return a + b;
    }

    // Overloaded method: Add three integers
    public int add(int a, int b, int c) {
        return a + b + c;
    }

    // Overloaded method: Add two doubles
    public double add(double a, double b) {
        return a + b;
    }
```

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```
    // Overloaded method: Concatenate two strings
    public String add(String a, String b) {
        return a + b;
    }
}

public class Main {
    public static void main(String[] args) {
        Calculator calc = new Calculator();

        // Call different add methods
        System.out.println("Sum of two integers: " + calc.add(5, 10)); // Calls add(int, int)
        System.out.println("Sum of three integers: " + calc.add(1, 2, 3)); // Calls add(int, int,
int)
        System.out.println("Sum of two doubles: " + calc.add(2.5, 3.5)); // Calls add(double,
double)
        System.out.println("Concatenation of strings: " + calc.add("Hello, ", "World!")); //
Calls add(String, String)
    }
}
```

Output:
Sum of two integers: 15
Sum of three integers: 6
Sum of two doubles: 6.0
Concatenation of strings: Hello, World!

b) Constructor Overloading

```
public class Employee {
    String name;
    int id;


    // Default constructor
    public Employee() {
        name = "Unknown";
        id = 0;
    }
```

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```
    // Parameterized constructor
    public Employee(String empName, int empId) {
        name = empName;
        id = empId;
    }


    public void display() {
        System.out.println("Name: " + name + ", ID: " + id);
    }


    public static void main(String[] args) {
        Employee emp1 = new Employee(); // Default constructor
        Employee emp2 = new Employee("John", 101); // Parameterized constructor


        emp1.display();
        emp2.display();
    }
}

Output:
Name: Unknown, ID: 0
Name: John, ID: 101
```

Conclusion: Hence, in this practical, we learned and executed the Method overloading and Constructor overloading in Java Successfully.